

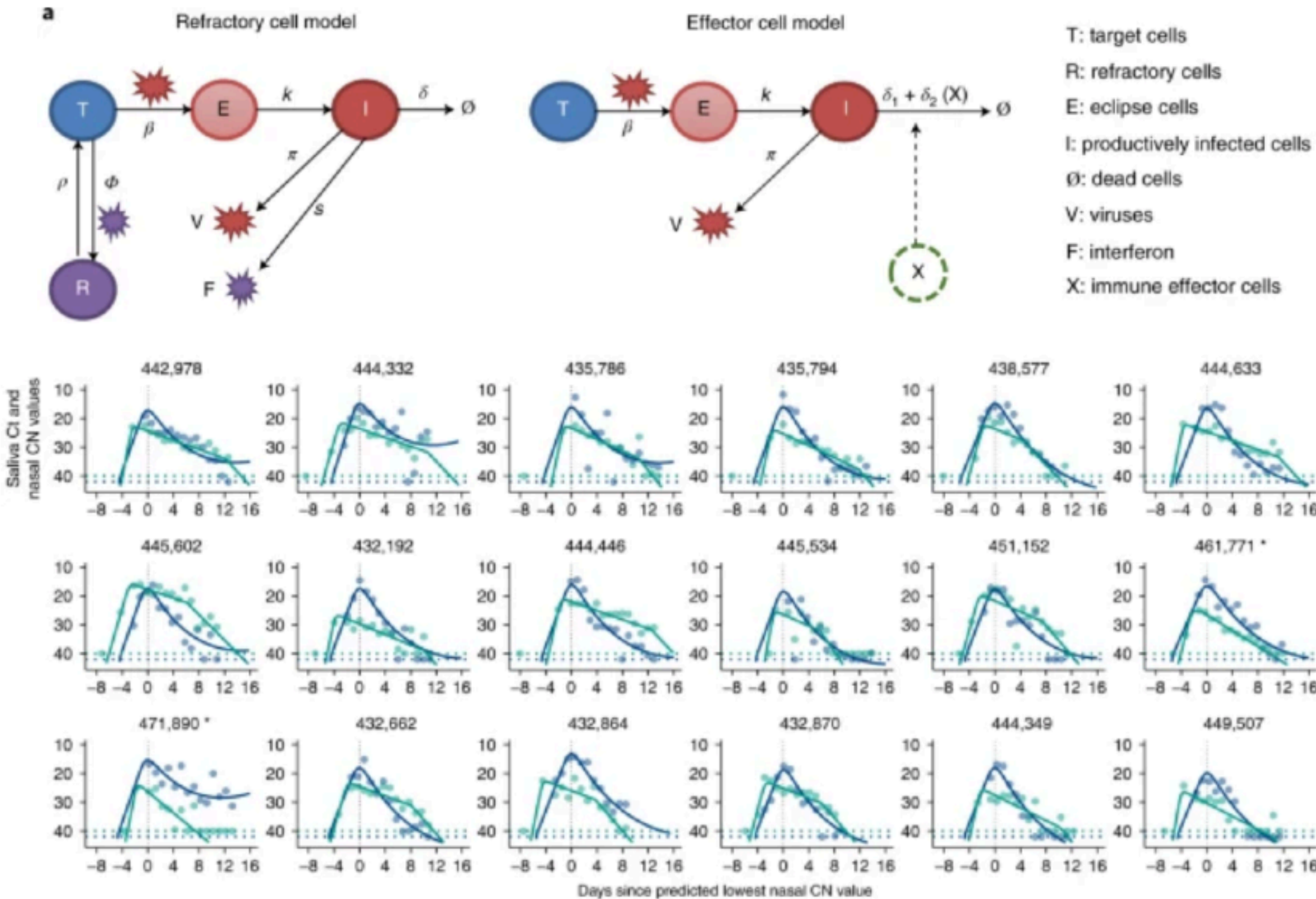
Modeling viral kinetics

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Daily longitudinal sampling of SARS-CoV-2 infection reveals substantial heterogeneity in infectiousness

Ruian Ke, Pamela P. Martinez, Rebecca L. Smith, Laura L. Gibson, Agha Mirza, Madison Conte, Nicholas Gallagher, Chun Huai Luo, Junko Jarrett, Ruifeng Zhou, Abigail Conte, Tongyu Liu, Mireille Farjo, Kimberly K. O. Walden, Gloria Rendon, Christopher J. Fields, Leyi Wang, Richard Fredrickson, Darci C. Edmonson, Melinda E. Baughman, Karen K. Chiu, Hannah Choi, Kevin R. Scardina, Shannon Bradley, Stacy L. Gloss, Crystal Reinhart, Jagadeesh Yedetore, Jessica Quicksall, Alyssa N. Owens, John Broach, Bruce Barton, Peter Lazar, William J. Heetderks, Matthew L. Robinson, Heba H. Mostafa, Yukari C. Manabe, Andrew Pekosz, David D. McManus & Christopher B. Brooke

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Viral kinetics of sequential SARS-CoV-2 infections

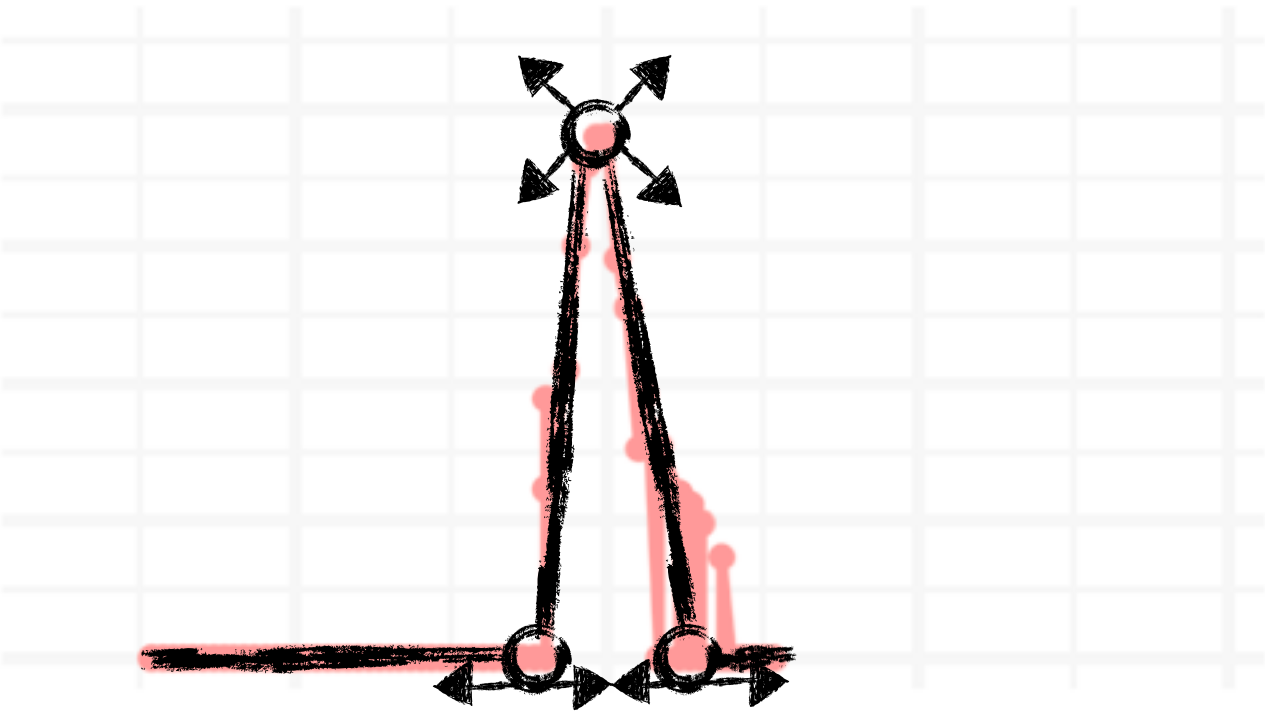
Stephen M. Kissler, James A. Hay, Joseph R. Fauver, Christina Mack, Caroline G. Tai, Deverick J. Anderson, David D. Ho, Nathan D. Grubaugh & Yonatan H. Grad

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$$\omega_{p[i]} = \text{Exp} \left[\beta_p + \beta_{p[c]} + \sum_a \beta_{p[a]} + \tau_p \eta_{p[i]} \right] \omega_p^*$$

$$\omega_{r[i]} = \text{Exp} \left[\beta_r + \beta_{r[c]} + \sum_a \beta_{r[a]} + \tau_r \eta_{r[i]} \right] \omega_r^*$$

$$\delta_{[i]} = \text{Exp} \left[\beta_\delta + \beta_{\delta[c]} + \sum_a \beta_{\delta[a]} + \tau_\delta \eta_{\delta[i]} \right] \delta^*$$



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Daily longitudinal mechanistic models of SARS-CoV-2 infection reveals substantial heterogeneity in infectiousness

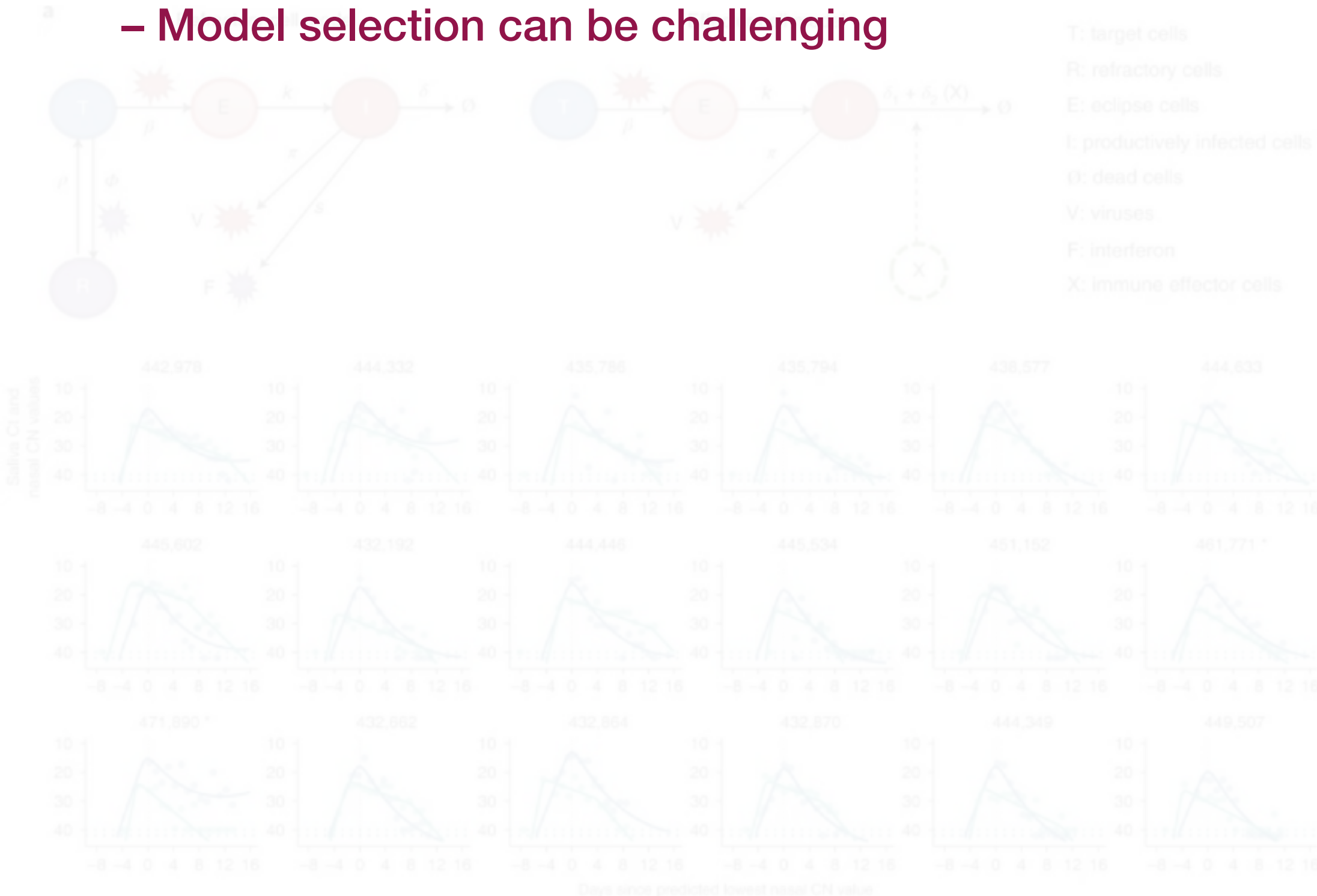
Ruian Ke, Pamela P. Martinez, Rebecca L. Smith, Laura L. Gibson, Agha Mirza, Madison Conte, Nicholas

+ Give insights about immunological mechanisms

+ Can be used to predict viral kinetic features (e.g., rebound)

– Can be challenging to fit/specify the parameters

– Model selection can be challenging



Viral kinetics Phenomenological models

Stephen M. Kissler, James A. Hay, Joseph R. Fauver, Christina Mack, Caroline G. Tai, Deverick J.

+ Provide epidemiologically relevant information with robust uncertainty measures

+ Quick/straightforward to fit, even to large datasets

– Provide less information about within-host mechanisms

– Can impose a too-rigid structure on the viral trajectory

